

Vikas Kumar Jha

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SUMMARY

Computer Science Engineering (Artificial Intelligence) student with strong foundations in Data Structures & Algorithms, Object-Oriented Programming and Operating Systems. Experienced in building software systems and AI-powered applications using C++, Python, LLMs, RAG, LangChain and LangGraph. Solved 500+ DSA problems with a peak LeetCode contest rating of 1711 (Top 15% globally).

EDUCATION

B.Tech in Computer Science Engineering (Artificial Intelligence) G.L. Bajaj Institute of Technology and Management, Greater Noida	2023 – 2027
Class XII (CBSE) Army Public School, Lucknow	2022
Class X (CBSE) Army Public School, Lucknow	2020

SKILLS

- **Languages:** C++, Python, Java
- **Core CS:** Data Structures & Algorithms, OOP, Operating Systems, DBMS
- **Generative AI:** LLMs, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering
- **Databases:** SQL, MongoDB
- **Tools:** Git, GitHub, VS Code

PROJECTS

Concurrent-KV — Multi-threaded In-Memory Key-Value Store C++, Sockets, Multithreading	Github
• Built an in-memory key-value store in C++ with a custom TCP server and thread-per-connection concurrency, using mutex synchronization for thread-safe access. • Implemented $O(1)$ LRU cache eviction using a hash map and doubly linked list, along with TTL-based expiration using lazy checks and a background sweep thread. • Added binary snapshot persistence for crash recovery and benchmarked concurrent throughput using a custom load-testing tool, identifying mutex contention beyond 16 concurrent clients.	
ZenTalk — Real-Time Chat Application React.js, Express.js, LangChain Team Project	Live Github
• Integrated Mistral Large (mistral-large-latest) to power an LLM-based spam detection system for classifying spam, phishing, and suspicious messages. • Developed a LangChain-based message moderation pipeline with confidence-based classification for identifying potentially harmful messages. • Implemented AI-powered chat summarization for community conversations, generating concise summaries from community messages.	
AskMe AI — AI Based Mock Interview Platform Python, Langchain, Langgraph, RAG	Github
• Built a multi-agent AI interview system using Gemini 2.5 Flash and DeepSeek-R1 for resume analysis, interview questioning, answer evaluation, and feedback generation. • Implemented RAG-based resume retrieval to extract relevant skills, experience, and projects and generate context-aware interview questions. • Designed a LangGraph workflow with LangChain to manage conversational interview flow, evaluate candidate responses, and generate technical and communication scores.	

CERTIFICATIONS

- Python Essentials 1 - Cisco Networking Academy
- Introduction to Generative AI - Google Cloud + Simplilearn
- Agentic AI Certified Foundations Associate — Oracle University

ACHIEVEMENTS

- Solved 500+ DSA problems with a peak LeetCode contest rating of 1711 (Top 15% globally).
- Winner of HackBuzz 2.0, organized by Vividhata Club, G.L. Bajaj Institute of Technology and Management.